

Assignment 1

Variational Free Energy and Importance Sampling

Overview

Given a reference measure Q_0 over trajectories and a cost functional $C(X)$, the optimal change of measure Q^* minimizes the free energy

$$\mathcal{F}(Q) = \mathbb{E}_Q[C(X)] + \alpha \text{KL}(Q \parallel Q_0),$$

where $\alpha > 0$ is the temperature parameter. The minimizer satisfies $Q^*(X) \propto Q_0(X) \exp(-C(X)/\alpha)$, and the optimal free energy has the dual form

$$\mathcal{F}(Q^*) = -\alpha \log \mathbb{E}_{Q_0}[\exp(-C(X)/\alpha)]. \quad (3.43)$$

Define the *desirability score* of a sample $X_i \sim Q_0$ as

$$r(i) = \exp(-C(X_i)/\alpha). \quad (3.45)$$

The free energy can then be estimated from N i.i.d. samples as $\hat{\mathcal{F}} = -\alpha \log(\frac{1}{N} \sum_i r(i))$. Sampling from Q^* itself requires reweighting the reference samples by their desirability scores (Algorithm 1).

In this assignment you will implement this method and study its behavior as a function of temperature and sample size.

Implementation

All code is in `variational/sampling.py`. Replace each `raise NotImplementedError` with your implementation.

1. Desirability scores (Eq. 3.45)

Implement `desirability_scores(C_values, alpha, xp)`.

Given N cost values and temperature α , compute the unnormalized importance weights $r(i)$. Use min-subtraction for numerical stability.

2. Free energy estimate (Eq. 3.43)

Implement `free_energy_estimate(C_values, alpha, xp)`.

Estimate $\mathcal{F}(Q^*)$ from the sample average of desirability scores. Use `desirability_scores()` and account for the min-subtraction offset.

3. Inverse CDF resampling (Algorithm 1)

Implement `inverse_cdf_resample(samples, rewards, n_resample, rng, xp)`.

Given reference samples $\{X_i\}$ and their desirability scores $\{r(i)\}$, draw new samples distributed according to Q^* using inverse CDF resampling.

Experiments

Boltzmann sampling on a 2D landscape

The reference measure R is uniform on $[0, 1]^2$, and the cost is a bowl with ripple structure: $C(x, y) = 8((x - 0.5)^2 + (y - 0.5)^2) + 0.8 \sin(3\pi x) \sin(3\pi y) + 0.5$. The optimal Q^* concentrates near the center where cost is lowest.

```
python examples/variational/boltzmann_sampling.py
python examples/variational/boltzmann_sampling.py --alpha 0.1
python examples/variational/boltzmann_sampling.py --alpha 10.0
```

Temperature sweep on a 1D double-well

The cost is $C(x) = (x^2 - 1)^2$, a symmetric double-well with minima at $x = \pm 1$ and a barrier at $x = 0$. The reference R is uniform on $[-2, 2]$. Five values of α show how the Boltzmann distribution transitions from bimodal to nearly uniform.

```
python examples/variational/alpha_sweep.py
```

1. Run `boltzmann_sampling.py` with $\alpha \in \{0.1, 1.0, 10.0\}$. How does the free energy convergence rate (in N) depend on α ? Relate your observation to the effective sample size $N_{\text{eff}} = (\sum_i w_i^2)^{-1}$, where $w_i = r(i) / \sum_j r(j)$.
2. For $\alpha = 0.1$ and $N = 1000$, what fraction of the weights w_i are numerically nonzero? What does this imply about the variance of the free energy estimate at low temperature?
3. Run `alpha_sweep.py` and examine the resampled histograms across the five temperature values. At what α does Q^* become effectively deterministic (concentrating on a single mode)?

Optional: GPU acceleration

Run the built-in benchmark and compare CPU vs GPU wall-clock time.

```
python examples/variational/boltzmann_sampling.py --benchmark
python examples/variational/boltzmann_sampling.py --benchmark --gpu
```

At what sample size does the GPU overhead become worthwhile?

Deliverables

- Completed `variational/sampling.py`.
- Convergence plots for three values of α .
- Written responses to experiment questions 1–3.